

Alif Azam

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Research Interest

Computational Fourier Optics · Physics-Informed Neural Networks
Scientific Machine Learning · High-Performance Computing

Skills

Languages	Python, C, Bash Scripting
Packages and Libraries	NumPy, Pandas, Matplotlib,
HPC Tools	Slurm, OpenMPI, OpenMP, CUDA, Git
Operating Systems	Linux, Windows

Education

B.Sc. Hons. in Physics

Shahjalal University of Science and Technology, Sylhet

January 2023-Present

Research Experience

Nonlinear Optics Laboratory

Department of Physics, SUST

December 2024 – Present

- Building a physics-informed spectral enhancement pipeline for label-free cell segmentation in phase-contrast microscopy, combining Fourier-domain quality assessment with PINN-based image restoration and adaptive bandpass filtering.

- Designed, built, and deployed a 7-node PXE-booted diskless HPC cluster — configured DHCP, TFTP, NFS, MUNGE, and Slurm for the master/compute-node boot and job-scheduling pipeline; later benchmarked the cluster using OpenMPI and OpenMP.

Publications

Conference Proceedings

1. Nripendro Biswas, Ahmed Rijvee, Munaem Ahmed Mahdi, **Alif Azam**, Md. Takoat Hossain, “*BSER – Bengali Speech Emotion Recognition using MFCC Features Based on SUBESCO and BanglaSER: A Comparative Study of Machine Learning Models.*” Proceedings of the 8th International Conference on Engineering Research, Innovation and Education (ICERIE 2025), Atlantis Press. DOI: 10.2991/978-94-6463-884-4_77

Presentations

1. "BSEER - Bengali Speech Emotion Recognition using MFCC Features Based on SUBESCO and BanglaSER: A Comparative Study of Machine Learning Models": Nripendro Biswas, Ahmed Rijvee, Munaem Ahmed Mahdi, **Alif Azam** and Md. Takoat Hossain. *International Conference on Engineering Research, Innovation and Education (ICERIE 2025)*.

Linguistic Skill

English (Fluent) · Bangla (Native)

Workshops/Courses

University of Colorado Boulder	link
Introduction to High-Performance and Parallel Computing	June 2026
Google	link
System Administration and IT Infrastructure Services	Jan 2025
Google Digital Marketing Course	July 2021
SoloLearn	
Python Core Course	Feb 2020

Extra-Curricular Activities

Assistant IT Secretary, Swapnotthan, SUST	May 2024–September 2024
Media and Outreach Team, SUST Autodrive	September 2024–November 2024
Volunteer, Sylhet Region, Bangladesh Physics Olympiad	December 2023
Typist, Chemistry Haters	August 2020–November 2020

Awards/Achievements

Chess Champion, Physics Indoor Tournament 2025	July 2025
Finisher, Protiki Marathon (5Km)	July 2025
Chess Runner-Up, Intra Department Indoor games, SUST Physics Society	December 2023
Uno Champion, Intra Department Indoor games, SUST Physics Society	December 2023